Enterprise Zero Trust Security Architecture

HL-CYB-001 | Four-week focused cybersecurity lab

A synthetic analyst needs report access, an engineer needs maintenance instructions and an unassigned user must be denied. This project designs a small isolated lab to demonstrate those decisions, remote revocation and recovery with evidence a reviewer can follow.

Current state: Aligned planning documents and repository checks exist. Lab implementation is **PLANNED**; operational controls are **NOT TESTED**; owner user acceptance testing (UAT) is **PENDING**. No production-impact or compliance claim is made.

1. Focused architecture

One Proxmox host runs four VMs: virtual OPNsense, one Windows Server with Active Directory/DNS/Group Policy, one Windows endpoint and one Linux HTTPS application server. Management, Corporate and Restricted are the three zones. Three synthetic standard users, three role groups and a separate privileged administrator support six application access cases and an access-revocation demonstration.

One simple lab certificate authority signs the application certificate. Tailscale alone exposes one named HTTPS resource in a separate lab tailnet. Two or three log sources, five representative events, one investigation, one repeatable PowerShell/Python operation and one measured application restore complete the lab.

The architecture explains flows, capacity and limits, including manual AD-to-application role mapping. The assignment owns ten FWR requirements, the four-week schedule and the single 100-point rubric. The guide groups practical steps into five phases while retaining separate stage A-L approvals.

2. Hardware and safety

The owner reports existing hardware and six ordered components. Ordered does not mean delivered, installed or validated. Hardware status and preparation records the distinction. Receipt, inspection, compatibility, current data/service inventory, backups/recovery and exact destructive-change approval precede installation. Additional hardware does not expand scope.

Start offline. No protected home/production trust, identities, credentials, routes, Microsoft 365, PE-DAP, customer data, corporate website testing or hosting integration is permitted. Independently isolated egress needs separate approval. Protected-network checks use configuration/passive review and lab-owned mocks, never probes of real protected systems.

3. Editable Lab Deliverable Templates

Five editable Word and Excel templates are available for the approved four-week lab: environment/architecture, identity/access implementation, monitoring/testing, troubleshooting and final submission. They contain planned prompts and empty answer areas, with all ten requirements and tests, mandatory subchecks and the approved 100-point rubric.

Keep working copies and raw evidence private. Fill them during separately approved lab stages and publish only reviewed summaries. The eight retired T5 shells remain in Git history; no historical result transfers to these templates.

4. Testing and evidence

Ten focused tests connect requirements, architecture, security controls, acceptance and future deliverables. All operational actual/evidence fields are empty. Acceptance requires completed mandatory subchecks and separate owner UAT; critical failures block acceptance regardless of score.

With Python 3.11+ and the existing pinned documentation environment:

```
python scripts/validate_docs.py --require-finalized
python scripts/validate_office_templates.py
python scripts/build_docs.py --check --rebuild-check
python -m unittest discover -s tests -v
git diff --check
```

These checks validate documentation, preservation and bounded publication risks; they do not test lab security. CI remains read-only, commit-pinned, without retained checkout credentials or lab access. Manual content/metadata/history review and rendered PDF inspection remain necessary.

5. Project documentation

The documentation library links Markdown sources and generated PDFs. Markdown is authoritative; task reports/addenda are Markdown only and frozen when finalized. Existing report PDFs remain byte-frozen archives.

- Security controls, decisions and risks explain safeguards and unresolved gates.
- T7 report records the five-file consolidation; T6 records specification alignment and T5 preserves original template validation.
- All task reports preserve engineering history; project final report remains a template.
- Glossary and official references explain terms and proposed commands.

Use focused task branches, reviewed public-safe commits/PRs, exact-head CI and intentional main synchronization under AGENTS.md. No background upload. Licensing and repository-protection settings remain owner decisions.

Next recommended task: T8, a read-only Stage A hardware and isolation readiness assessment, after explicit owner approval of assets, scope, existing tools, validation, private evidence and recovery/stop conditions. No installation or later stage is included.